

Local Decoder.

David Ponarovsky

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Introduction

Today:

- ▶ Proof.

1. Local Decoders.

Our Code

Code

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1 \Delta_2$.

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$ -depolarized channel:

- ▶ There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- ▶ There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift-rule and the flip upon majority are local decoders.

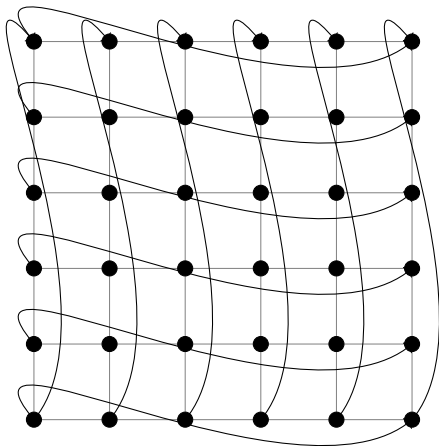


Figure: On the left is the Toric Graph. On the right are cross and face checks.

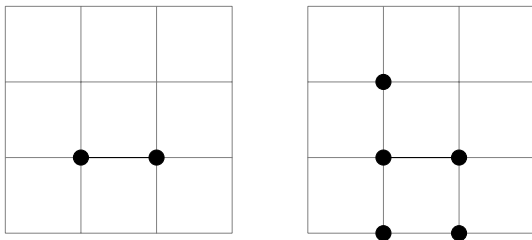


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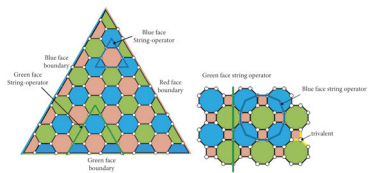


Figure: 2d color code

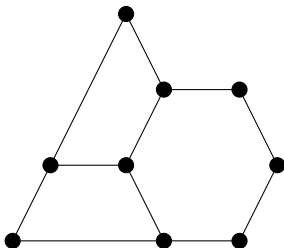


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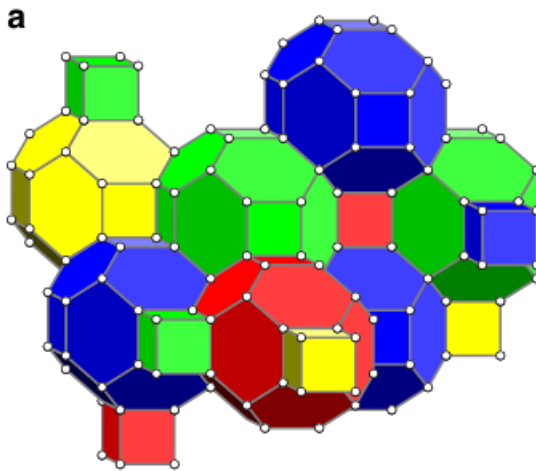


Figure 1 | Representations of the lattice geometry of the gauge color code. (a) A 3D visualization of the lattice geometry, where the three-dimensional cells of the lattice are four colorable, that is, every cell can be assigned a color. (b) The gauge color code of linear size $L = 5$ drawn in the dual picture. In this representation, each cell is a tetrahedron, and the four colors are used to color the faces of the tetrahedra such that no two faces of a given tetrahedron have the same color.